Proof. As $\dim \operatorname{End} V_i = (\dim V_i)^2$, Theorem 3.5.4 implies that $\dim A - \dim \operatorname{Rad}(A) = \sum_i \dim \operatorname{End} V_i = \sum_i (\dim V_i)^2$. As $\dim \operatorname{Rad}(A) \geq 0$, $\sum_i (\dim V_i)^2 \leq \dim A$. $\square$

Example 3.5.6. 1. Let $A = k[x]/(x^n)$. This algebra has a unique irreducible representation, which is a 1-dimensional space k, in which x acts by zero. So the radical $\operatorname{Rad}(A)$ is the ideal (x).

2. Let A be the algebra of upper triangular $n \times n$ matrices. It is easy to check that the irreducible representations of A are V_i, $i = 1, \ldots, n$, which are 1-dimensional, and any matrix x acts by x_{ii}. So the radical $\operatorname{Rad}(A)$ is the ideal of strictly upper triangular matrices (as it is a nilpotent ideal and contains the radical). A similar result holds for block-triangular matrices.

Definition 3.5.7. A finite dimensional algebra A is said to be **semisimple**[4] if $\operatorname{Rad}(A) = 0$.

Proposition 3.5.8. *For a finite dimensional algebra A, the following are equivalent:*

(1) *A is semisimple.*

(2) *$\sum_i (\dim V_i)^2 = \dim A$, where the V_i's are the irreducible representations of A.*

(3) *$A \cong \bigoplus_i \operatorname{Mat}_{d_i}(k)$ for some d_i.*

(4) *Any finite dimensional representation of A is completely reducible (that is, isomorphic to a direct sum of irreducible representations).*

(5) *A is a completely reducible representation of A.*

Proof. As $\dim A - \dim \operatorname{Rad}(A) = \sum_i (\dim V_i)^2$, clearly $\dim A = \sum_i (\dim V_i)^2$ if and only if $\operatorname{Rad}(A) = 0$. Thus, (1) $\Leftrightarrow$ (2).

By Theorem 3.5.4, if $\operatorname{Rad}(A) = 0$, then clearly $A \cong \bigoplus_i \operatorname{Mat}_{d_i}(k)$ for $d_i = \dim V_i$. Thus, (1) $\Rightarrow$ (3).

Next, (3) $\Rightarrow$ (4) by Theorem 3.3.1. Clearly (4) $\Rightarrow$ (5).

[4]In particular, the algebra $A = 0$ is semisimple, although it is not simple. Every representation of this algebra is zero, so it does not have any irreducible or indecomposable representations. It is, nevertheless, a direct sum of an (empty) collection of matrix algebras.